

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 22
Design of FIR Filters Using the Windowing Method

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept & Principle of Windowing

1.1 The Truncation Problem

An ideal brick-wall filter (e.g. ideal lowpass with cutoff ω_c) has a frequency response $H_d(e^{j\omega})$. Its impulse response $h_d[n]$ is infinite in duration (IIR) and non-causal:

$$h_{d,LP}[n] = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega = \frac{\sin(\omega_c n)}{\pi n}, \quad -\infty < n < \infty \quad (1)$$

To realize a physical causal FIR filter of length N , we:

1. Shift the center of symmetry to $\tau = \alpha = \frac{N-1}{2}$ to ensure causality.
2. Multiply $h_d[n]$ pointwise by a finite-duration window sequence $w[n]$ ($0 \leq n \leq N-1$):

$$\mathbf{h}[\mathbf{n}] = \mathbf{h}_d[\mathbf{n}] \cdot \mathbf{w}[\mathbf{n}], \quad 0 \leq n \leq N-1 \quad (2)$$

1.2 Frequency-Domain Convolution & Gibbs Phenomenon

By the DTFT modulation property, time-domain multiplication corresponds to periodic convolution in frequency:

$$H(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) \circledast W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta \quad (3)$$

- **Mainlobe Width of $W(e^{j\omega})$:** Spreads out the ideal vertical transition into a gradual transition band $\Delta\omega$.
- **Sidelobes of $W(e^{j\omega})$:** Cause oscillations (ringing) in the passband and stopband.
- **Gibbs Phenomenon:** Abrupt rectangular truncation results in a fixed 8.95% overshoot (≈ -21 dB peak sidelobe) near the discontinuity, which **cannot be reduced by increasing N** . Smoother, tapered windows are necessary to suppress sidelobes.

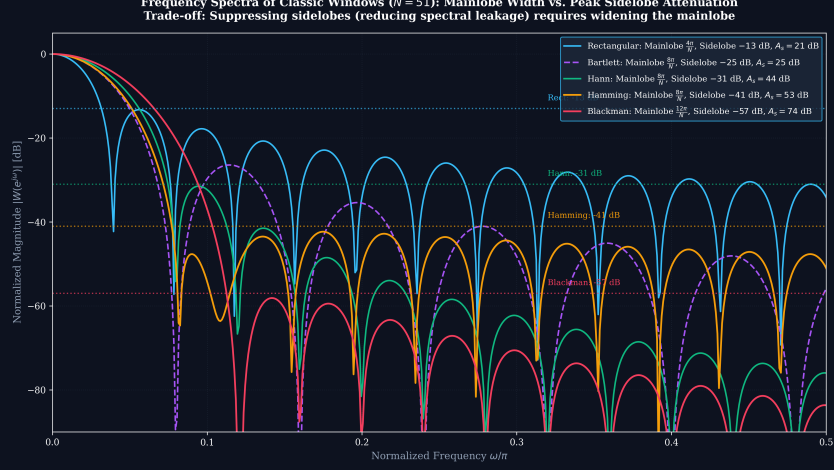


Figure 1: Frequency spectra of classic window functions showing the fundamental trade-off: narrower mainlobe (faster transition) vs. lower sidelobes (higher stopband attenuation).

2 Master Window Summary Table (Formulas & Specs)

Table 1: Characteristics of Standard FIR Filter Window Functions

Window Name	Time-Domain Equation $w[n]$ ($0 \leq n \leq N-1$)	Transition Width $\Delta\omega$	Peak Sidelobe	Min. Attenuation A_s
Rectangular	$w[n] = 1$	$\frac{4\pi}{N}$ (or $\frac{1.8\pi}{N}$)	-13 dB	21 dB
Bartlett (Triangular)	$w[n] = 1 - \frac{2 n-(N-1)/2 }{N-1}$	$\frac{8\pi}{N}$ (or $\frac{6.1\pi}{N}$)	-27 dB	25 dB
Hann (Hanning)	$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$	$\frac{8\pi}{N}$ (or $\frac{6.2\pi}{N}$)	-32 dB	44 dB
Hamming	$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$	$\frac{8\pi}{N}$ (or $\frac{6.6\pi}{N}$)	-43 dB	53 dB
Blackman	$w[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$	$\frac{12\pi}{N}$ (or $\frac{11\pi}{N}$)	-58 dB	74 dB

3 Ideal Shifted Impulse Response Formulations

All ideal impulse responses must be delayed by $\tau = \alpha = \frac{N-1}{2}$:

Table 2: Closed-Form Formulas for Ideal Shifted Filters $h_d[n]$

Filter Type	For $n \neq \tau$	Center Tap $n = \tau$
Lowpass (LPF)	$h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = \frac{\omega_c}{\pi}$
Highpass (HPF)	$h_d[n] = -\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = 1 - \frac{\omega_c}{\pi}$
Bandpass (BPF)	$h_d[n] = \frac{\sin(\omega_{c2}(n-\tau)) - \sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = \frac{\omega_{c2} - \omega_{c1}}{\pi}$
Bandstop (BSF)	$h_d[n] = \frac{\sin(\omega_{c1}(n-\tau)) - \sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}$

4 5-Step FIR Filter Window Design Procedure

1. **Identify Required Attenuation A_s :** From stopband specification $A_s = -20 \log_{10}(\delta_s)$ dB, select the appropriate window from Table 1 that meets or exceeds A_s .
2. **Calculate Filter Length N :** From the transition bandwidth $\Delta\omega = |\omega_s - \omega_p|$, calculate N using the window's transition width formula. Round up to the **next odd integer** to ensure Type 1 linear phase.
3. **Determine Cutoff Frequency ω_c :** Average the passband and stopband edge frequencies:

$$\omega_c = \frac{\omega_p + \omega_s}{2} \quad (4)$$

4. **Compute Ideal Impulse Response $h_d[n]$:** Evaluate $h_d[n]$ for $n = 0, 1, \dots, N-1$ using Table 2.
5. **Apply Window:** Compute final filter coefficients: $h[n] = h_d[n] \cdot w[n]$.

5 Standard Solved Exam Example

Problem: Design a linear-phase FIR lowpass filter with a cutoff frequency $\omega_c = 0.5\pi$ rad/sample and filter length $N = 7$ using a **Hamming window**.

Solution:

Step 1: Parameters & Center of Symmetry Length $N = 7$ (Odd), center $\tau = \frac{7-1}{2} = 3$. Cutoff $\omega_c = 0.5\pi$.

Step 2: Ideal Impulse Response $h_d[n]$

$$h_d[n] = \frac{\sin(0.5\pi(n-3))}{\pi(n-3)}, \quad n \neq 3; \quad h_d[3] = \frac{0.5\pi}{\pi} = 0.5 \quad (5)$$

Evaluating for $n = 0, 1, 2, 3$:

- $n = 3$: $h_d[3] = 0.5000$
- $n = 2$: $h_d[2] = \frac{\sin(0.5\pi(-1))}{\pi(-1)} = \frac{-1}{-\pi} = \frac{1}{\pi} \approx \mathbf{0.3183}$
- $n = 1$: $h_d[1] = \frac{\sin(0.5\pi(-2))}{\pi(-2)} = \frac{\sin(-\pi)}{-2\pi} = \mathbf{0.0000}$
- $n = 0$: $h_d[0] = \frac{\sin(0.5\pi(-3))}{\pi(-3)} = \frac{1}{-3\pi} \approx \mathbf{-0.1061}$

By symmetry about $\tau = 3$: $h_d[4] = h_d[2] = 0.3183$, $h_d[5] = h_d[1] = 0.0$, $h_d[6] = h_d[0] = -0.1061$.

Step 3: Hamming Window Weights $w[n]$

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{6}\right) = 0.54 - 0.46 \cos\left(\frac{\pi n}{3}\right) \quad (6)$$

Evaluating for $n = 0, 1, 2, 3$:

- $n = 3$: $w[3] = 0.54 - 0.46 \cos(\pi) = 0.54 + 0.46 = \mathbf{1.0000}$
- $n = 2$: $w[2] = 0.54 - 0.46 \cos(2\pi/3) = 0.54 - 0.46(-0.5) = 0.54 + 0.23 = \mathbf{0.7700}$
- $n = 1$: $w[1] = 0.54 - 0.46 \cos(\pi/3) = 0.54 - 0.46(0.5) = 0.54 - 0.23 = \mathbf{0.3100}$

- $n = 0$: $w[0] = 0.54 - 0.46 \cos(0) = 0.54 - 0.46 = \mathbf{0.0800}$

Step 4: Final Impulse Response Coefficients $h[n] = h_d[n] \cdot w[n]$

- $h[3] = h_d[3] \cdot w[3] = 0.5000 \times 1.0000 = \mathbf{0.5000}$
- $h[2] = h[4] = h_d[2] \cdot w[2] = 0.31831 \times 0.7700 = \mathbf{0.2451}$
- $h[1] = h[5] = h_d[1] \cdot w[1] = 0.0000 \times 0.3100 = \mathbf{0.0000}$
- $h[0] = h[6] = h_d[0] \cdot w[0] = -0.10610 \times 0.0800 = \mathbf{-0.0085}$

The final designed filter is:

$$\mathbf{h[n]} = \{-0.0085, 0.0000, 0.2451, \underset{\uparrow}{0.5000}, 0.2451, 0.0000, -0.0085\} \quad (7)$$

Symmetry check: $h[n] = h[6 - n]$ is satisfied. The filter is a Type 1 linear-phase lowpass filter.

6 Essential Exam Checklist & Traps

- ☐ **Center Tap Formula:** For LPF, $h_d[\tau] = \omega_c/\pi$. Do NOT attempt $\frac{\sin(0)}{0}$ without taking the limit!
- ☐ **Highpass Inversion:** For HPF, $h_d[\tau] = 1 - \omega_c/\pi$. All other taps are the negative of the LPF taps.
- ☐ **Window Selection Rule:** If the specification requires $A_s = 50$ dB, a Hann window ($A_s = 44$ dB) is insufficient. You **must select Hamming** ($A_s = 53$ dB).
- ☐ **Argument in Window Cosine:** Remember the denominator is $N-1$, not N : $\cos\left(\frac{2\pi n}{N-1}\right)$.